

Space Exploration

Just as the last big boom in technological innovation came at the hands of space exploration, it isn't surprising that space is proving to be the next testing ground for 3D printing as well. With NASA and private business' partnering to install 3D printing facilities in the International Space Station, dreamers believe it will be a critical step towards colonisation in the long run.

So far everything that humanity has sent into space has been made on Earth, it may no longer be the case once the goal of 3D printing in space is realised. Objects launched from Earth are not only meant to function in zero gravity but also withstand the journey to space which involves launch vibrations and g-forces. But by manufacturing objects directly in zero-gravity a whole new range of functionality, flexibility and utility may be achieved.

By taking advantage of the zero-gravity or micro-gravity environment, engineers and designers can create objects that would not be functional on Earth but perfectly efficient in space. In addition to which, the on-demand nature of 3D printing would put to ease any



3D printing in space is already a reality and marks a big step towards space colonisation.

concerns of spare parts, repairs, costs and even food. But so far these attempts are in the experimental stages, the creation of a microgravity friendly 3D printer has itself been a struggle as it needs to be compact, secure, safe, launch ready and materials adaptable.

After numerous failed prototypes a solution has been achieved and is in the process of implementation. Designed by "Made in Space", the 3D printer has been approved by NASA engineers and will be sent to space in 2014 to manufacture the first object outside of Earth. Plans to launch an updated advanced version of the next generation 3D printer by "Made in Space" is already in play with a scheduled 2015 launch date. 